

— PORTFOLIO CASE STUDY

AI Part Prioritization Framework

Systemizing sourcing triage under ambiguity

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V2.0

01 — PROBLEM

If everything is important, what is actually urgent?

Sourcing engineers in complex manufacturing are routinely handed portfolios of 50 to 100 parts and asked to drive impact. There is rarely a structured method to determine where to allocate effort, which risks are existential versus noise, or which cost opportunities represent real leverage.

Prioritization defaults to reactive, meeting-driven, and personality-influenced, not system-driven.

"Why are you working on that part first?" A question that rarely had a data-backed answer.

02 — INSIGHT

Production continuity first. Not spend alone.

A \$50K part that stops a line is more urgent than a \$2M part with diversified supply and stable delivery. The right system must be explainable, surface operational risk, and produce defensible action categories without relying on escalation pressure or institutional memory.

Supply continuity failures in aerospace are catastrophic and difficult to reverse. Cost opportunity, while meaningful, is a second-order priority.

A weighted scoring engine across four dimensions

PRIORITY SCORE FORMULA

Score = ((Production Impact x 0.35) + (Supply Risk x 0.30)

+ (Cost Exposure x 0.20) + (Instability x 0.15)) / 5 x 100

Weights are configurable. Production First (Default): 35/30/20/15 | Cost Down: 15/20/40/25 | Supply Resilience: 25/40/20/15 | Custom: user-defined

35%

PRODUCTION IMPACT

— CTB status

— Line-stop risk

— Safety stock (days)

30%

SUPPLY RISK

— Single-source exposure

— Supplier count

— Geographic concentration

— Special process requirements

20%

COST EXPOSURE

— Annual spend

— Should-cost delta (%)

15%

INSTABILITY

— On-time delivery %

— Quality issues

— Chronic open status

Default weights reflect a production-continuity-first philosophy. All weights are configurable in the SETTINGS tab. Switch modes to reprioritize the portfolio based on current business objective without rebuilding the model.

04 — V2 UPGRADES

Four new capabilities built on the V1 foundation

Dynamic Weight Configuration	A dedicated SETTINGS tab allows users to switch between three preset scoring modes: Production First, Cost Down, and Supply Resilience, or define custom weights. The entire ranking updates instantly without touching any formulas.
Reason Codes	Each ranked part now displays a plain-language reason code in the OUTPUT tab explaining why it scored where it did. Example: "CTB, Single Source, OTD Critical, High Cost Delta." Eliminates the need to scan across eight columns in leadership meetings.
Should-Cost Flag	Parts missing should-cost delta data are flagged with a warning in the OUTPUT tab. Surfaces incomplete data before sourcing action is taken rather than silently treating missing values as zero.

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Risk vs. Opportunity Matrix	A dedicated MATRIX tab plots all 60 parts across Supply Risk (X) and Cost Exposure (Y). Four quadrants, Act Now, Cost Opportunity, Risk Watch, and Monitor, give an instant visual overview of portfolio composition and sourcing priorities.
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05 — FRAMEWORK CONTROLS

Three features that make the model ops-ready for day-to-day sourcing workflows

Data Confidence Flag	Each part is rated H/M/L for input reliability. Low-confidence parts surface a warning before action is taken. The tool informs, the sourcing manager decides.
Escalation Override	Leadership-escalated parts receive a minimum score of 70 and forced action category without corrupting the underlying calculated score. Political context stays visible without contaminating model integrity.
AI Rationale Layer	Top 10 ranked parts are passed through a structured Claude prompt generating a 3-sentence business rationale and action recommendation in plain language, ready for leadership reviews. AI does not create scores or facts — it only rewrites computed drivers into a narrative. Scoring and action categories are fully deterministic.

06 — OUTPUT EXAMPLE

From qualitative noise to structured tradeoffs

Instead of "Part A feels urgent," the system produces a scored, ranked, and explainable output with reason codes automatically generated from the underlying data:

P-1040 — High-Temp Injector Plate (synthetic)

90 / 100

DIMENSION SCORES

PRODUCTION IMPACT

5.0

SUPPLY RISK

4.0

COST EXPOSURE

4.0

INSTABILITY

5.0

ACTION CATEGORY

ESCALATE — Supplier Crisis + Production Risk

CTB, Single Source, OTD Critical, High Cost Delta, \$1M+ Spend, Low Safety Stock

Immediate action: dual-source qualification and cost recovery negotiation.

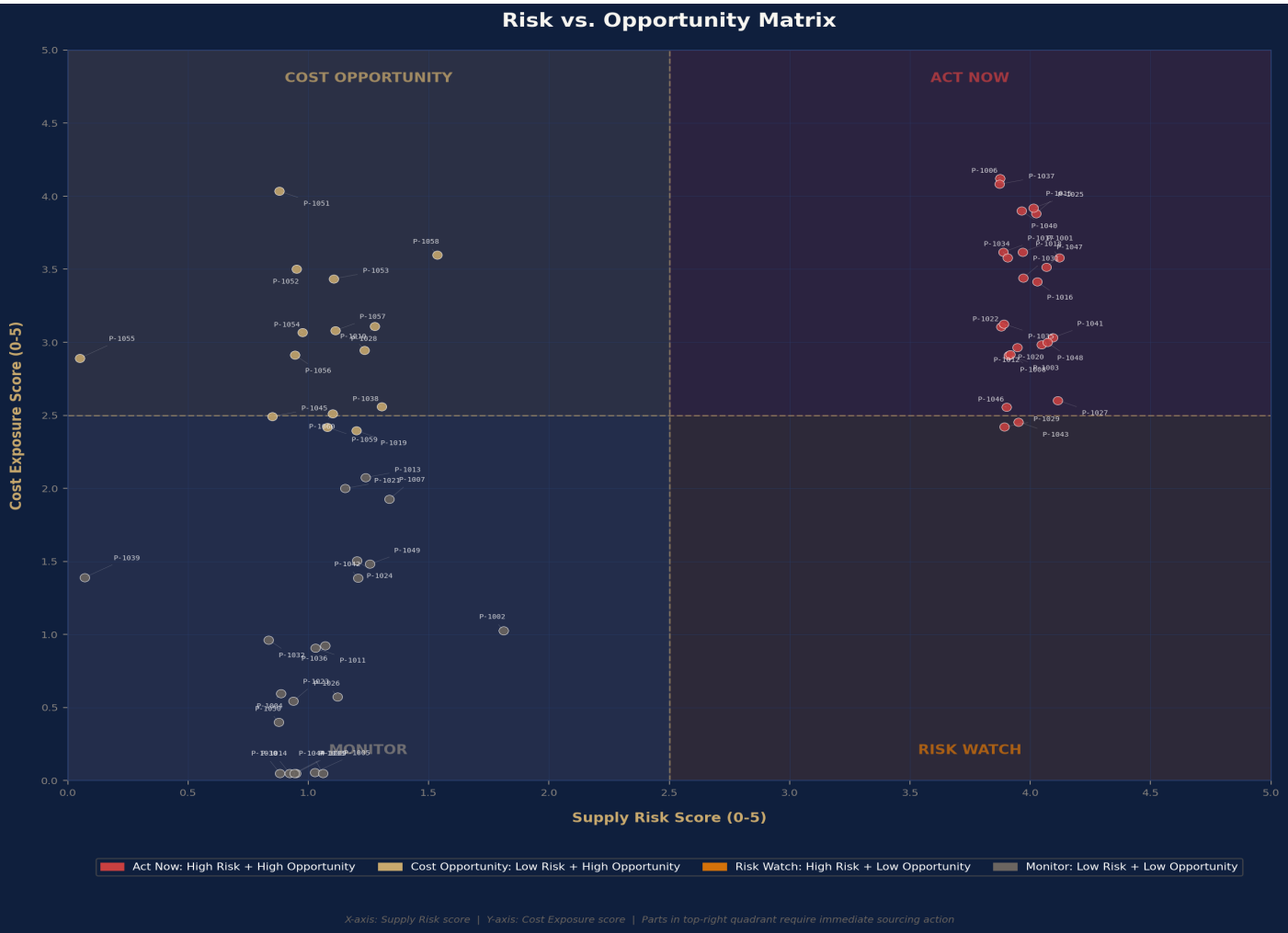
All parts, part numbers, and values shown above are synthetic and illustrative only.

OUTPUT Tab — 60-Part Ranked Priority List with Reason Codes

Rank	Part Number	Part Description	Priority Score	Production Impact	Supply Risk	Cost Exposure	Instability	Action Category	Rank Key	Data Warning	Should-Cost Flag	Reason Code
1	P-1040	Inconel Injector Plate	90.0	5	4	4	5	ESCALATE — Supplier Crisis + Production Risk	90.041			CTB Single Source OTD Critical High Cost Delta \$1M+ Spend Low Safety Stock
2	P-1037	Titanium Engine Mount	90.0	5	4	4	5	ESCALATE — Supplier Crisis + Production Risk	90.038			CTB Single Source OTD Critical High Cost Delta \$1M+ Spend Low Safety Stock
3	P-1025	Inconel Turbine Blade	90.0	5	4	4	5	ESCALATE — Supplier Crisis + Production Risk	90.026			CTB Single Source OTD Critical High Cost Delta \$1M+ Spend Low Safety Stock
4	P-1015	Inconel Exhaust Nozzle	90.0	5	4	4	5	ESCALATE — Supplier Crisis + Production Risk	90.016			CTB Single Source OTD Critical High Cost Delta \$1M+ Spend Low Safety Stock
5	P-1006	Avionics Mounting Plate	90.0	5	4	4	5	ESCALATE — Supplier Crisis + Production Risk	90.007			CTB Single Source OTD Critical High Cost Delta \$1M+ Spend Low Safety Stock
6	P-1047	Inconel Nozzle Extension	88.0	5	4	3.5	5	ESCALATE — Supplier Crisis + Production Risk	88.048			CTB Single Source OTD Critical High Cost Delta \$1M+ Spend Low Safety Stock
7	P-1034	Steel Combustion Liner	88.0	5	4	3.5	5	ESCALATE — Supplier Crisis + Production Risk	88.035			CTB Single Source OTD Critical High Cost Delta Low Safety Stock
8	P-1031	Inconel Heat Shield	88.0	5	4	3.5	5	ESCALATE — Supplier Crisis + Production Risk	88.032			CTB Single Source OTD Critical High Cost Delta \$1M+ Spend Low Safety Stock
9	P-1018	Inconel Valve Body	88.0	5	4	3.5	5	ESCALATE — Supplier Crisis + Production Risk	88.019			CTB Single Source OTD Critical High Cost Delta Low Safety Stock
10	P-1016	Titanium Thrust Chamber	88.0	5	4	3.5	5	ESCALATE — Supplier Crisis + Production Risk	88.017			CTB Single Source OTD Critical High Cost Delta \$1M+ Spend Low Safety Stock
11	P-1001	Titanium Bracket Assembly	88.0	5	4	3.5	5	ESCALATE — Supplier Crisis + Production Risk	88.002			CTB Single Source OTD Critical \$1M+ Spend Low Safety Stock
12	P-1017	Carbon Fiber Nose Cone	85.0	5	4	3.5	4	ESCALATE — Supplier Crisis + Production Risk	85.018			CTB Single Source OTD Poor \$1M+ Spend Low Safety Stock
13	P-1003	Carbon Fiber Panel - Aft	83.0	5	4	3	4	ESCALATE — Supplier Crisis + Production Risk	83.004			CTB Single Source OTD Poor High Cost Delta Low Safety Stock
14	P-1048	Carbon Fiber Interstage	69.0	3	4	3	4	Strategic RFQ — Cost Down + Risk	69.049			Single Source OTD Poor High Cost Delta Low Safety Stock
15	P-1020	Steel Pressure Vessel	69.0	3	4	3	4	Strategic RFQ — Cost Down + Risk	69.021			Single Source OTD Poor Low Safety Stock
16	P-1008	Composite Duct Assembly	69.0	3	4	3	4	Strategic RFQ — Cost Down + Risk	69.009			Single Source OTD Poor Low Safety Stock
17	P-1028	Steel Hydraulic Manifold	66.5	5	1.25	3	4	ESCALATE — Supplier Crisis + Production Risk	66.529			CTB OTD Poor High Cost Delta Low Safety Stock

RATIONALE Tab — AI Generated Output (Top 10)

Part Number	Part Description	Priority Score	Action Category	AI Rationale
P-1040	Inconel Injector Plate	90	ESCALATE — Supplier Crisis + Production Risk	P-1040 scored 90/100 primarily because it combines maximum production criticality with severe supplier instability: it is a critical-to-build component with only 2 days of safety stock and high line-stop risk, meaning any supply disruption converts directly into halted production within 48 hours. The single-source supply base with a special process requirement eliminates short-term substitution options, while a 62% OTD rate, active quality issues, and a chronic open action log confirm this supplier is in sustained underperformance — not an isolated incident. Compounding the operational risk, the \$1.35M annual spend carries a 25% should-cost delta, meaning the business is paying a significant premium above fair value for a part that is already failing delivery and quality commitments. ACTION: Convene an executive-level supplier crisis review within 5 business days, impose a formal recovery plan with 30/60/90-day OTD and quality gates, and immediately initiate qualification of a secondary source capable of meeting the Inconel special process requirement.
P-1037	Titanium Engine Mount	90	ESCALATE — Supplier Crisis + Production Risk	P-1037 scored 90/100 because it presents an immediate production continuity threat: as a critical-to-build component with only 1 day of safety stock and high line-stop risk, the current 60% OTD rate means the line is statistically overdue for a supply-driven stoppage with virtually no buffer to absorb it. The single-source structure combined with a titanium special process requirement means there is no qualified fallback supplier, making every missed delivery a direct escalation event with no mitigation path available inside the current supply base. At \$1.95M annual spend with a 29% should-cost delta, the business is paying nearly \$565K above fair value annually to a supplier that is simultaneously failing on delivery, carrying active quality issues, and holding a chronic open action log. ACTION: Issue a formal supplier default notice within 48 hours, require a recovery plan with weekly OTD and quality checkpoints, and fast-track qualification of an alternative titanium special-process supplier given the 1-day safety stock exposure.
P-1025	Inconel Turbine Blade	90	ESCALATE — Supplier Crisis + Production Risk	P-1025 scored 90/100 because it combines zero supply chain redundancy with the worst OTD rate in this part family: at 59% on-time delivery against a 2-day safety stock buffer, the production line is absorbing near-constant late deliveries with almost no cushion, and a single missed shipment can trigger a line stop on a critical-to-build component. The single-source structure with an Inconel special process requirement means there is no qualified alternate supplier that can be activated in a crisis, locking the business into a supplier that is simultaneously failing delivery, generating active quality escapes, and carrying a chronic open action log with no resolution. At \$1.65M annual spend with a 31% should-cost delta — the highest cost gap among the 90-point parts — the business is absorbing roughly \$510K annually above fair market value with no performance leverage, indicating the supplier has fully priced in their sole-source position. ACTION: Place P-1025 on a formal supplier watch list with a 30-day cure notice, mandate weekly delivery and quality performance reviews, and immediately fund a secondary source qualification for Inconel turbine blade special process capability as the highest-cost-exposure item in the escalation portfolio.



07 — ACTION LOGIC

Score profile drives action, not total score alone

The same overall score can result from very different underlying conditions. Action categories are assigned by profile. Supplier Crisis takes precedence over Dual Source escalation when both conditions are met.

CONDITION	ACTION CATEGORY
Leadership Escalation = Y	ESCALATE — Active Leadership
Production Impact 4+ and Instability 4+	ESCALATE — Supplier Crisis + Production Risk
Production Impact 4+ and Supply Risk 3.5+	ESCALATE — Dual Source + Eng Review
Production Impact 4+	Dual Source Priority
Supply Risk 3+ and Cost Exposure 3+	Strategic RFQ — Cost Down + Risk

Supply Risk 4+	Supply Risk Mitigation
Cost Exposure 4+	Cost Down / RFQ
Instability 3+	Supplier Performance Review
Priority Score 40 or below	Monitor

08 — BROADER CONTEXT

One tool in a sourcing operations suite

This framework is the first in a series of structured sourcing tools being built to address the most common pain points in manufacturing procurement. Each tool is designed to stand alone, solve a specific operational problem, and produce outputs that are defensible in a leadership setting.

Upcoming tools in development: RFQ Tracker, Should-Cost Model Assistant, Supplier Risk Scorecard, Make vs. Buy Decision Framework.

09 — CLOSING NOTE

Built to structure sourcing decisions under pressure. Intentionally simple in structure, rigorous in logic. The goal is clarity, repeatability, and leverage, not complexity. The model surfaces the data. The sourcing manager makes the call.

Excel-native decision support tool, built iteratively across two versions. 60-part synthetic dataset, structure validated against real workflow constraints. 7-tab workbook. Production-continuity-first philosophy.